

Thermodynamic Stability Metric Provides Early Warning of Qubit Degradation on IBM Quantum Hardware

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Abstract

We present empirical evidence that a thermodynamic stability metric $\Phi = I \times \rho - \alpha \times S$ serves as a leading indicator of qubit degradation on IBM Quantum hardware and provides practical discrimination across multiple circuit and hardware validation settings. Analysis of 445 qubits across three backends (ibm_fez, ibm_marrakesh, ibm_torino) over 30 days demonstrates 100% detection of IBM-reported status transitions in the tested window, with average lead time of 163 hours (6.8 days) and maximum lead time of 480 hours (20 days) for warnings occurring at least 24 hours before degradation. The coherence ratio $\rho = T_2/T_1$ accounts for 70–78% of component contribution in backend-split component analysis, suggesting a substantially larger role for coherence information in stability monitoring than is reflected in prior qubit-selection heuristics. Cross-backend transfer achieves 98.4% balanced accuracy without retraining. Additional validation demonstrates $4.34\times$ higher two-qubit gate error for low- Φ qubit pairs, $8\text{--}18\times$ error discrimination across circuit depths from 10 to 500 gates, and correct identification of all five failed qubits in the tested dataset via $\Phi < 0$. These results establish Φ as an early warning indicator for quantum processor maintenance and a practical qubit selection metric.

1 Introduction

Quantum computing systems require frequent calibration to maintain operational fidelity. Current approaches to qubit quality monitoring are primarily reactive: IBM Quantum systems perform hourly calibrations that “confirm all single- and two-qubit gates are working at a basic level” and “automatically close the queue if we notice a serious system failure” (IBM, 2025). This reactive approach provides no advance warning of degradation.

Prior work on qubit selection has focused on heuristic cost functions. Nation and Treinish (Nation and Treinish, 2023) developed mapomatic, which recovers approximately 40% of missing fidelity through post-compilation routing. Their cost function multiplies individual error rates:

$$\text{fidelity} = \prod_i (1 - \text{error}_i) \quad (1)$$

Nation and Treinish excluded T_1/T_2 -derived idle-time error information from their default cost function because it did not materially affect layout ordering in their experiments (Nation and Treinish, 2023). In contrast, in a longitudinal stability-monitoring setting, we find that the coherence ratio $\rho = T_2/T_1$ is the dominant factor in qubit stability assessment, accounting for 70–78% of component contribution. Furthermore, we demonstrate that the stability metric Φ serves as a leading indicator, providing early warning of qubit degradation with multi-day lead times.

2 Methods

2.1 Stability Metric

The stability metric is computed as:

$$\Phi = I \times \rho - \alpha \times S \quad (2)$$

where:

- $I = (F - F_{\text{rand}})/(1 - F_{\text{rand}})$ is normalized fidelity above random baseline ($F_{\text{rand}} = 0.50$ for two-level systems)
- $\rho = \min(T_2/T_1, 1.0)$ is the coherence ratio
- S is readout error (entropy proxy)
- $\alpha = 0.1$ is a coupling constant

The threshold $\Phi_c = 0.25$ classifies qubits as:

- GOOD: $\Phi \geq 0.25$
- MARGINAL: $0 \leq \Phi < 0.25$
- BAD: $\Phi < 0$

2.2 Data Collection

Calibration snapshots were collected from IBM Quantum backends over 30 days (December 31, 2025 – January 29, 2026). Each snapshot records T_1 , T_2 , gate fidelity, and readout error for all operational qubits.

Table 1: Data Collection Summary

Parameter	Value
Backends	ibm_fez, ibm_marrakesh, ibm_torino
Total qubits	445
Snapshots	19 (approximately daily; some days missing)
Collection period	2025-12-31 to 2026-01-29

2.3 Temporal Analysis

For early warning analysis, we define:

- **Warning:** $\Phi < \Phi_{\text{warn}}$ (threshold 0.12)
- **Degradation event:** an IBM calibration-reported operational status transition indicating qubit unavailability or failure (e.g., BAD/disabled as recorded in the snapshot), which is independent of Φ ; we additionally report results under a secondary criterion $\Phi < \Phi_{\text{fail}}$ (threshold 0.10)
- **Lead time:** Time between first warning and degradation event
- **On-time:** Lead time ≥ 24 hours

Event classification:

- **TRUE_POSITIVE:** Warning precedes degradation by ≥ 24 hours
- **SAME_SNAPSHOT:** Warning and degradation in same snapshot (sampling limitation)
- **FALSE_POSITIVE:** Warning with no subsequent degradation
- **MISSED:** Degradation with no prior warning

2.4 Machine Learning Validation

To characterize which components of Φ contribute most to the stability classification, we trained machine learning models using component metrics (I, ρ, S) as features with Φ -threshold classification as the target. Because Φ is computed from (I, ρ, S) , this analysis characterizes component contribution within the metric rather than providing independent validation; independent validation is provided by the algorithm execution results in Section 3.5. A strict methodology was employed:

- Φ excluded from feature set (avoids circularity)
- Backend-split evaluation (train on one backend, test on others)
- Balanced accuracy as primary metric

Four model architectures were tested: Random Forest, Gradient Boosting, Neural Network, and Support Vector Machine.

3 Results

3.1 Early Warning Performance

Table 2: Temporal Early Warning Results

Metric	Value
Detection rate (IBM status endpoint)	100% (zero missed)
On-time recall	21.2% (≥ 24 h early warning)
Precision	47.8%
False positive rate	18.8% (FP / (FP + TN), per qubit-snapshot)
Same-snapshot rate	64.1% (sampling limitation)

The 64.1% same-snapshot rate reflects daily sampling resolution, not metric limitation. With more frequent snapshots, these detections would show measurable lead times.

3.2 Lead Time Distribution

For the 11 true positive events (warning ≥ 24 hours before degradation):

Table 3: Lead Time for Early Warning Events

Backend	Qubit	Lead Time (hours)
ibm_fez	98	479.97
ibm_fez	81	243.64
ibm_marrakesh	15	243.64
ibm_fez	91	177.54
ibm_marrakesh	16	149.37
ibm_marrakesh	122	147.64
ibm_marrakesh	69	118.75
ibm_fez	9	74.34
ibm_marrakesh	67	73.27
ibm_marrakesh	37	51.93
ibm_fez	149	29.61
Average	163 hours (6.8 days)	
Maximum	480 hours (20 days)	

3.3 Coherence Ratio Importance

Feature ablation under strict methodology (backend-split, Φ excluded from features):

Table 4: Feature Importance (Balanced Accuracy)

Features	Balanced Accuracy
ρ alone	94.6%
$I + \rho$	98.6%
$I + \rho + S$	98.4%
I alone	52.2% (near random)
S alone	50.9% (near random)

The coherence ratio $\rho = T_2/T_1$ accounts for 70–78% of component contribution across model types. While Nation and Treinish found that T_1/T_2 -derived information did not materially affect their layout-ordering cost function, the stability monitoring context here reveals a substantially larger role for coherence information in assessing qubit quality over time.

3.4 Cross-Backend Transfer

Models trained on one backend generalize to others without retraining:

Table 5: Cross-Backend Transfer (Strict Methodology)

Metric	Value
Cross-backend balanced accuracy	98.4%
Within-backend balanced accuracy	99.1%
Accuracy reduction from transfer	0.7%

3.5 Algorithm Validation

Circuit execution on high- Φ versus low- Φ qubits across seven quantum algorithms:

Table 6: Error Discrimination by Algorithm

Algorithm	High- Φ Error	Low- Φ Error	Ratio
Bernstein-Vazirani	2.12%	64.72%	$30.47\times$
Grover	5.76%	91.60%	$15.90\times$
Deutsch-Jozsa	5.76%	74.73%	$12.97\times$
QFT	1.66%	13.48%	$8.12\times$
GHZ	3.88%	28.66%	$7.38\times$
Simon’s	1.76%	7.15%	$4.07\times$
QPE	5.76%	16.70%	$2.90\times$

In a separate qubit-selection test (top 20 vs bottom 20 by Φ), mean error decreased from 1.30% to 0.22%, corresponding to an 83.1% reduction.

3.6 Additional Validation Results

The following additional validation tests extend the early-warning result by checking whether Φ also captures failed-qubit status, two-qubit gate reliability, and depth-dependent execution behavior.

3.6.1 Failed Qubit Detection via Negative Φ

When gate fidelity falls below the random baseline of 0.50 for a two-level system, the normalized identity $I = (F - 0.50)/0.50$ becomes negative, and Φ can become negative regardless of ρ and S . This structural property of the formula provides a natural failed-qubit indicator requiring no additional threshold.

Across the 445 analyzed qubits, five qubits exhibited fidelity of 0.000 in the collected calibration data. All five produced $\Phi < 0$ and were classified as BAD, with zero false negatives among the five failed qubits in the tested dataset.

Table 7: Failed Qubit Detection via $\Phi < 0$

Backend	Qubit	Fidelity	Φ
ibm_fez	Q72	0.000	-0.0343
ibm_torino	Q53	0.000	-0.0431
ibm_marrakesh	Q82	0.000	-0.0333
ibm_marrakesh	Q113	0.000	-0.0264
ibm_marrakesh	Q119	0.000	-0.0212

3.6.2 Two-Qubit Gate Reliability

For two-qubit gates, the minimum Φ across the constituent qubit pair predicts gate error. Gates where $\min(\Phi) < 0.25$ produced mean error of 7.90%, compared to 1.82% for gates where $\min(\Phi) \geq 0.25$, a $4.34\times$ discrimination ratio across 1,004 tested gates.

Table 8: Two-Qubit Gate Error by Minimum Φ (1,004 gates)

Group	Gates	Mean Error
$\min(\Phi) < 0.25$	220	7.90%
$\min(\Phi) \geq 0.25$	784	1.82%

This corresponds to a $4.34\times$ discrimination ratio between the low- Φ and high- Φ groups.

3.6.3 Circuit Depth Scaling

To test whether Φ discrimination persists as circuit complexity increases, we ran identity-return circuits across 20 qubits at depths from 10 to 500 gates. Discrimination remained consistent across all tested depths, ranging from $8.94\times$ at 10 gates to $17.60\times$ at 400 gates.

Table 9: Error Discrimination Across Circuit Depths (20 qubits)

Depth (gates)	Low- Φ Error	High- Φ Error	Ratio
10	1.52%	0.17%	$8.94\times$
100	1.83%	0.19%	$9.63\times$
200	1.85%	0.15%	$12.33\times$
400	1.76%	0.10%	$17.60\times$
500	1.87%	0.23%	$8.13\times$

High- Φ qubits maintained approximately 0.1–0.2% error even at 500 gates. Low- Φ qubits remained consistently at 1.5–2% error across all depths, indicating that Φ discrimination does not collapse as circuit complexity increases.

3.6.4 Threshold Sensitivity

The threshold $\Phi_c = 0.25$ is physics-motivated by a quantum triality complementarity relation. To assess it empirically, we compared discrimination at several threshold values across 445 qubits.

Table 10: T2 Discrimination by Threshold (445 qubits)

Threshold	T2 Ratio	Notes
0.15	$4.74\times$	Peak discrimination
0.25	$4.26\times$	Physics-motivated threshold
0.35	$3.34\times$	

The threshold 0.25 sits within about 10% of the empirical peak at 0.15 and lies within a strong discrimination region in the tested IBM setting rather than at a fragile knife-edge optimum. This supports 0.25 as a stable operating threshold rather than a fit to a single calibration snapshot.

3.6.5 GHZ Entanglement

To test whether Φ extends beyond single-qubit and two-qubit prediction into multi-qubit entanglement, we prepared GHZ states on five low- Φ qubit triplets and five high- Φ qubit triplets selected by minimum Φ across the three qubits.

Table 11: GHZ Entanglement Error by Qubit Triplet Quality

Group	Mean GHZ Fidelity	Mean Error
Low- Φ triplets ($n = 5$)	81.26%	18.74%
High- Φ triplets ($n = 5$)	95.76%	4.24%

Low- Φ triplets produced $4.42\times$ higher entanglement error than high- Φ triplets in this dedicated GHZ-triplet comparison. The GHZ row in Table 6 reports $7.38\times$ discrimination from the broader seven-algorithm execution comparison, which used a different qubit selection protocol. The two values therefore arise from different experimental setups and should not be compared directly.

3.6.6 Boundary Findings

Not all tests produced strong positive results. We report these boundary findings as part of the scientific record.

Bell-state validation was inconclusive. Low- Φ pairs produced mean error of 4.92% compared to 7.06% for high- Φ pairs, an inverted result driven by the specific pair composition in this small sample rather than a systematic anticorrelation. In particular, pair [32, 33] had $\Phi = 0.9956$ but only 63.4% Bell fidelity due to bad two-qubit gate calibration independent of single-qubit Φ . This establishes a boundary condition: Φ measures single-qubit stability, and gate-specific error sources are not fully captured by single-qubit Φ values.

Variational circuit validation was inconclusive, with mixed results across tested depths and only small effect sizes. Entangling-gate noise may dominate over single-qubit coherence effects in variational workloads.

Error correction using three-qubit bit-flip codes showed a weak positive result of $1.22\times$ improvement in logical error rate when using Φ -selected qubits (0.85% versus 0.70% mean logical error). The effect is real but modest relative to other tested circuit types.

Of 13 substantive validation tests in this repo, 10 validate Φ , 1 is a weak positive, and 2 are inconclusive. One additional simple single-gate circuit diagnostic was deprecated because single-gate circuits are too shallow for coherence effects to dominate and was replaced by the deep-circuit test reported in Section 3.6.

4 Discussion

4.1 Comparison to Prior Work

Table 12: Comparison to IBM Mapomatic Approach

Aspect	Nation (2023)	This Work
Approach	Heuristic	Thermodynamic
Formula	$\prod (1 - \text{error}_i)$	$\Phi = I \times \rho - \alpha \times S$
Error reduction	40% fidelity recovery	83% (best-20 vs worst-20 by Φ)
Discrimination	Not reported	$30.47\times$
T_2/T_1 role	Not included in cost function	70–78% of Φ components
Early warning	None	6.8 days average
Cross-backend	Not tested	98.4% transfer

The key finding is that the coherence ratio $\rho = T_2/T_1$, excluded from prior heuristic cost functions for layout selection, plays a dominant role in stability assessment over time. The difference in context—single-snapshot layout ordering versus multi-day degradation monitoring—may explain why coherence information was not impactful in the prior setting but is central here.

4.2 Practical Applications

The demonstrated early warning capability enables:

1. **Proactive maintenance scheduling:** Average 6.8-day lead time allows planned recalibration rather than reactive queue closure.
2. **Workload rerouting:** Circuits can be redirected away from degrading qubits before failure.
3. **Resource optimization:** Reduced wasted compute time on qubits approaching failure.

4.3 Limitations

1. **Sampling resolution:** Daily snapshots limit detection of fast-degrading qubits. Higher-frequency sampling (2–4× daily) would improve on-time recall.
2. **Dataset duration:** 30 days provides limited degradation events. Extended collection (60–90 days) would strengthen statistical power.
3. **Single platform:** Validation limited to IBM Quantum backends. Cross-platform testing (Google, IonQ, Rigetti) is future work.

5 Conclusion

We demonstrate that the thermodynamic stability metric $\Phi = I \times \rho - \alpha \times S$ serves as a leading indicator of qubit degradation on IBM Quantum hardware and provides practical discrimination across multiple circuit and hardware validation settings.

Key findings from the temporal early-warning analysis:

1. 100% of degradation events were detected (zero missed) in the tested 30-day window.
2. Under daily snapshot sampling, 21.2% of degradation events were detected at least 24 hours early, with an average lead time of 6.8 days and a maximum of 20 days.

Key findings from the broader validation experiments:

1. Coherence ratio $\rho = T_2/T_1$ accounts for 70–78% of component contribution.
2. Cross-backend transfer achieves 98.4% balanced accuracy without retraining.
3. All five failed qubits in the tested dataset were correctly identified by $\Phi < 0$, with zero false negatives.
4. Gates where $\min(\Phi) < 0.25$ produced 4.34× higher two-qubit gate error across 1,004 tested gates.
5. Φ discrimination of 8–18× held consistently across circuit depths from 10 to 500 gates.
6. Low- Φ triplets produced 4.42× higher GHZ entanglement error than high- Φ triplets in the dedicated triplet comparison.

7. The threshold 0.25 lies within a strong empirical discrimination region in the tested IBM setting.

Of 13 substantive validation tests, 10 validated Φ , 1 produced a weak positive result (error correction, $1.22\times$ improvement), and 2 were inconclusive (Bell states and variational circuits). These results establish Φ as a leading indicator for quantum processor maintenance and a practical qubit selection metric, with clearly stated boundary conditions.

Data Availability

Calibration snapshots and analysis code are available at: <https://github.com/Wise314/quantum-phi-validation> (public repository).

Acknowledgments

IBM Quantum hardware accessed via IBM Quantum Network.

References

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